

ASP.NET Core Web Portal – Requirements Specification

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Table of Contents

1. Web Portal Overview
 2. Stakeholder Needs (Web-Specific)
 3. Functional Requirements (FR5–FR8)
 4. Admin User Roles
 5. Web Portal Pages
 6. Non-Functional Requirements (Web-Specific)
 7. Integration Requirements
 8. Technology Stack
 9. Database Tables (Web-Specific)
 10. UI Wireframe Concepts
 11. Phase 1 Reflection
 12. Next Steps – Phase 2
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1. Web Portal Overview

1.1 Purpose

The ASP.NET Core Web Portal serves as the central management dashboard for Mangojane Funeral Parlor administrators. It provides real-time operational visibility and control over:

- Booking processing and approval.
- Resource allocation, including hearses, drivers and equipment.
- Policy and payment management.
- Daily scheduling and dispatch reporting.

The portal is intended to improve the management of funeral operations by replacing manual processes with a centralised digital system.

1.2 Target Users

User Type	Description
Manager	Full operational access to approve bookings, manage resources and view reports.
Dispatcher	Assigns assets, views schedules and updates operational statuses.
Front-Desk Staff	Views bookings, records payments and searches policy information.
Management	Monitors operational information, reports and system activity.

1.3 Portal Objectives

Objective	Description
Real-Time Visibility	Administrators can view booking requests shortly after clients submit them.
Conflict Prevention	The system helps prevent double-booking of hearses and equipment.
Efficient Processing	Administrators can quickly approve, reject or update booking requests.
Policy Management	Staff can search and verify burial society policy information.
Operational Reporting	Staff can generate and view daily dispatch schedules and reports.

2. Stakeholder Needs (Web-Specific)

Based on stakeholder requirements gathering and project discussions, the following needs were identified for the web portal:

Need	Importance	How the Web Portal Addresses It
Eliminate paper diary double-bookings	CRITICAL	Calendar view and resource conflict detection.
Quickly verify policy status	HIGH	Searchable policy records.
Real-time service schedule visibility	HIGH	Dashboard displaying booking and scheduling information.
Easy payment logging	MEDIUM	Payment recording and payment history functionality.
Staff coordination	MEDIUM	Daily dispatch schedules and reports.
Data privacy	CRITICAL	Authentication and role-based access control.

3. Functional Requirements (FR5–FR8)

FR5: Admin Booking Approval and Rejection

Requirement	Description	Priority
FR5.1	Display incoming booking requests in a sortable and searchable table.	MUST HAVE
FR5.2	Display the client name, selected package, preferred date and booking status.	MUST HAVE
FR5.3	Allow authorised administrators to approve, reject or update booking requests.	MUST HAVE
FR5.4	Update booking status information through the shared Supabase backend.	MUST HAVE
FR5.5	Record the user responsible for approving or rejecting a booking and the date and time of the action.	SHOULD HAVE

User Flow

1. An administrator logs into the web portal.
2. The administrator views pending bookings on the dashboard.
3. The administrator selects a booking request.
4. The administrator reviews the client and booking details.
5. The administrator approves or rejects the booking.
6. The booking status is updated in the shared database.
7. The updated status becomes available to the mobile application.

FR6: Asset and Resource Scheduling

Requirement	Description	Priority
FR6.1	Assign hearses, drivers and equipment to approved bookings.	MUST HAVE
FR6.2	Provide a visual schedule or calendar for resource allocation.	MUST HAVE
FR6.3	Prevent double-booking of the same resource for conflicting dates or times.	MUST HAVE
FR6.4	Display the availability of resources.	MUST HAVE
FR6.5	Allow authorised users to override a scheduling conflict where appropriate and record a reason.	SHOULD HAVE

Asset Types

Asset Type	Description	Example Quantity
Hearse	Vehicle used to transport the deceased.	2
Family Car	Vehicle used for family transportation.	2
Tent	Tent used during funeral or burial services.	3
Lowering Equipment	Equipment used during burial services.	2
Chairs	Seating provided for funeral services.	50

Asset Type	Description	Example Quantity
Staff	Drivers and operational staff.	4

FR7: Policy and Payment Management

Requirement	Description	Priority
FR7.1	Search policy records using a policy number or policyholder name.	MUST HAVE
FR7.2	View policy status and payment history.	MUST HAVE
FR7.3	Record premium or policy payments.	MUST HAVE
FR7.4	Generate a payment receipt in PDF format.	SHOULD HAVE
FR7.5	Display outstanding payment balances.	COULD HAVE

Policy Information

Field	Description
Policy Number	Unique identifier for the policy.
Policyholder Name	Full name of the policyholder.
Policy Type	Type of funeral policy or package.
Status	Active, Lapsed or Closed.
Payment History	Payment date, amount and payment method.
Outstanding Balance	Amount still owed, where applicable.

Note: Live banking and payment gateway integration are outside the scope of this prototype. Payment information will be recorded or simulated for academic demonstration purposes.

FR8: Daily Dispatch Reports

Requirement	Description	Priority
FR8.1	Generate daily dispatch reports for authorised staff.	SHOULD HAVE
FR8.2	Include hearse assignments, driver schedules and funeral or cemetery locations.	SHOULD HAVE
FR8.3	Allow reports to be viewed digitally or printed.	SHOULD HAVE
FR8.4	Allow reports to be exported to PDF format.	SHOULD HAVE
FR8.5	Allow reports to be shared electronically with staff.	COULD HAVE

4. Admin User Roles

4.1 Role Definitions

Role	Access Level	Permissions
Super Admin	Full	All CRUD operations, user management and system settings.
Manager	High	Approve or reject bookings, manage assets and view reports.
Dispatcher	Medium	Assign resources, view schedules and update operational information.
Front-Desk	Limited	View bookings, record payments and search policies.
View Only	Restricted	View selected reports and information without editing.

4.2 Permission Matrix

Function	Super Admin	Manager	Dispatcher	Front-Desk	View Only
View Bookings	Yes	Yes	Yes	Yes	Yes
Approve/Reject Bookings	Yes	Yes	No	No	No
Assign Assets	Yes	Yes	Yes	No	No
Manage Policies	Yes	Yes	No	Yes	No
Log Payments	Yes	Yes	No	Yes	No
Generate Reports	Yes	Yes	No	No	Yes
Manage Users	Yes	No	No	No	No
System Settings	Yes	No	No	No	No

5. Web Portal Pages

5.1 Login Page

The Login Page shall allow authorised employees to access the web portal.

The page shall include:

- Email or username field.
 - Password field.
 - Login button.
 - Error message for invalid credentials.
 - Secure authentication through the selected authentication system.
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5.2 Dashboard Page

The Dashboard shall provide a summary of important operational information.

The dashboard may display:

- Total bookings.
 - Pending booking requests.
 - Approved bookings.
 - Upcoming funerals.
 - Available hearses.
 - Scheduled resources.
 - Recent payments.
 - Daily operational schedule.
-

5.3 Booking Management Page

The Booking Management Page shall allow authorised users to:

- View all booking requests.
- Search and filter bookings.
- View booking details.
- Approve bookings.
- Reject bookings.
- Update booking statuses.

Booking information may include:

- Client name.
 - Contact information.
 - Selected package.
 - Preferred funeral date.
 - Funeral location.
 - Booking status.
-

5.4 Booking Details Page

The Booking Details Page shall provide complete information about an individual booking.

The administrator shall be able to:

- Review client details.
- View the selected funeral package.
- View the requested funeral date.

- Assign available resources.
 - Approve or reject the booking.
 - Update the booking status.
-

5.5 Asset Scheduling Page

The Asset Scheduling Page shall provide a calendar or schedule view for managing resources.

The page shall allow authorised users to:

- View available hearses and equipment.
 - Assign assets to bookings.
 - View existing assignments.
 - Detect scheduling conflicts.
 - Prevent double-booking.
-

5.6 Policy Management Page

The Policy Management Page shall allow authorised users to:

- Search for policy records.
 - View policyholder information.
 - View policy status.
 - View payment history.
 - Update relevant policy information where authorised.
-

5.7 Payment Management Page

The Payment Management Page shall allow authorised users to:

- Record payments.
 - View payment history.
 - Search payment records.
 - View outstanding balances.
 - Generate payment receipts where implemented.
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5.8 Daily Dispatch Reports Page

The Daily Dispatch Reports Page shall allow authorised users to:

- Select a report date.
 - View scheduled funerals.
 - View assigned hearses.
 - View assigned drivers.
 - View funeral locations.
 - Generate a dispatch report.
 - Print or export the report to PDF.
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6. Non-Functional Requirements (Web-Specific)

6.1 Performance

NFR1: The web portal should load major pages within a reasonable response time under normal operating conditions.

NFR2: Booking and scheduling information should be updated promptly after changes are made.

6.2 Security

NFR3: The system shall require authentication before users can access protected administrative functions.

NFR4: The system shall implement role-based access control.

NFR5: Users shall only be able to access functions appropriate to their assigned role.

NFR6: Sensitive user and operational information shall be protected through appropriate security mechanisms.

6.3 Usability

NFR7: The web portal shall provide a clear and user-friendly interface.

NFR8: Navigation shall allow users to easily access major portal functions.

NFR9: The system shall provide clear success, warning and error messages.

6.4 Reliability

NFR10: The system should maintain consistent booking and scheduling information.

NFR11: The system shall prevent duplicate or conflicting resource assignments.

NFR12: Important system actions should be recorded where appropriate for accountability.

6.5 Maintainability

NFR13: The application should be developed using a structured and maintainable architecture.

NFR14: The code should be organised to allow future updates and additional features.

7. Integration Requirements

The ASP.NET Core Web Portal shall integrate with the shared Supabase backend.

The integration shall allow the web portal to:

- Retrieve booking requests submitted through the Flutter mobile application.
- Retrieve client information.
- Update booking statuses.
- Store resource assignments.
- Retrieve policy information.
- Record payment information.
- Synchronise relevant data between the mobile application and web portal.

The web portal and mobile application shall use the shared backend as a central source of system information.

8. Technology Stack

Component	Technology
Web Framework	ASP.NET Core
Programming Language	C#
Frontend	HTML, CSS and JavaScript
Database / Backend	Supabase

Component	Technology
Authentication	Supabase Authentication
Data Synchronisation	Supabase Real-Time Services
Development Environment	Visual Studio / Visual Studio Code
Version Control	Git and GitHub

9. Database Tables (Web-Specific)

The web portal will interact with the shared database through the following main tables.

9.1 Users

Field	Description
UserID	Unique user identifier.
FullName	Full name of the user.
Email	User email address.
Role	Assigned system role.
CreatedAt	Date the account was created.

9.2 Bookings

Field	Description
BookingID	Unique booking identifier.
ClientID	Linked client identifier.
PackageID	Selected funeral package.
FuneralDate	Planned funeral date.
Location	Funeral or service location.
Status	Current booking status.
CreatedAt	Date the booking was created.

9.3 Assets

Field	Description
AssetID	Unique asset identifier.
AssetName	Name or description of the asset.
AssetType	Type of resource.
AvailabilityStatus	Current availability status.

9.4 Asset Assignments

Field	Description
AssignmentID	Unique assignment identifier.
BookingID	Related booking.
AssetID	Assigned asset.
AssignedDate	Date of assignment.
StartTime	Scheduled start time.
EndTime	Scheduled end time.

9.5 Policies

Field	Description
PolicyID	Unique policy identifier.
PolicyNumber	Policy reference number.
PolicyholderName	Name of the policyholder.
PolicyType	Type of policy.
Status	Policy status.

9.6 Payments

Field	Description
PaymentID	Unique payment identifier.
PolicyID	Related policy.
Amount	Payment amount.
PaymentDate	Date of payment.
PaymentMethod	Method used for payment.
RecordedBy	User who recorded the payment.

10. UI Wireframe Concepts

The web portal interface will follow a simple administrative dashboard structure.

10.1 Main Layout

The main portal layout will include:

- Top navigation area.
- Side navigation menu.

- Main content area.
- User profile or logout section.

Proposed Navigation Menu

- Dashboard
 - Bookings
 - Asset Scheduling
 - Policies
 - Payments
 - Reports
 - Users
 - Settings
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10.2 Dashboard Concept

The dashboard will display summary cards for:

- Pending Bookings.
- Approved Bookings.
- Upcoming Funerals.
- Available Assets.
- Recent Payments.

A table may also display recent booking requests.

10.3 Booking Management Concept

The Booking Management Page will use a table containing:

Client Package Funeral Date Status Action

The Action column may allow authorised users to:

- View details.
 - Approve.
 - Reject.
 - Update the booking.
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10.4 Asset Scheduling Concept

The Asset Scheduling Page will provide a calendar or schedule view showing:

- Dates.
- Bookings.
- Assigned hearses.
- Assigned equipment.
- Resource availability.

Conflicting assignments should be clearly indicated to prevent double-booking.

11. Phase 1 Reflection

During Phase 1, the main focus was on understanding the operational requirements of Mangojane Funeral Parlor and translating those requirements into web-based system functionality.

The requirements analysis identified several challenges associated with manual operational processes, including booking conflicts, difficulty managing resources, limited visibility of schedules and challenges related to managing policy and payment information.

The proposed ASP.NET Core Web Portal will provide a centralised administrative platform to address these challenges. The portal will allow authorised users to process booking requests, schedule funeral resources, manage policy and payment information and generate operational reports.

The requirements defined during this phase will guide the design and development activities in the next phase of the project.

12. Next Steps – Phase 2

The following activities are planned for Phase 2:

1. Design the database structure and relationships.
 2. Create detailed user interface wireframes.
 3. Set up the ASP.NET Core project.
 4. Configure the connection to the Supabase backend.
 5. Implement authentication and role-based access.
 6. Develop the booking management module.
 7. Develop the asset scheduling functionality.
 8. Develop the policy and payment management module.
 9. Develop the daily dispatch reporting functionality.
 10. Test integration between the web portal and shared backend.
 11. Perform functional testing of the implemented features.
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